The Klenow fragment of DNA polymerase I of *E. coli* was the first DNA polymerase structure solved. Unlike the complete enzyme, which possesses exonuclease activity directed toward each end of a growing DNA strand, the Klenow fragment contains only 3′→5′ exonuclease activity, which is activated when a recently added base pair encounters the exonuclease region. The structure of the Klenow fragment, including the relative positions of its polymerase and exonuclease regions as well as the boundaries of its "right-hand" regions, are shown in Figure 1.

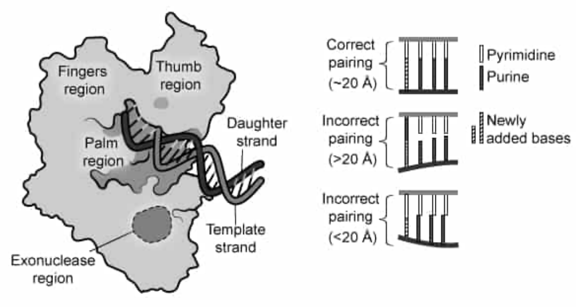


Figure 1 The Klenow fragment of DNA polymerase I (Note: The polymerase and exonuclease regions are separated by 35 angstroms.)

A team of researchers used the Klenow fragment to synthesize experimental oligonucleotides labelled with ^{32}P. These oligonucleotides contained either the binding sequence for a transcription factor called nuclear factor-kappa B (NF-κB) or a mutated sequence of the same length that lacks the binding site. The oligonucleotide probes were added to nuclei isolated from liver cells previously treated with the natural product curcumin or with a control solution.

After incubation, the samples underwent polyacrylamide gel electrophoresis. NF-κB proteins are dimers that may consist of various combinations of subunits, and antibodies that bind to specific NF-κB subunits were applied to some samples prior to electrophoresis. Typical band patterns and intensities from the electrophoretic mobility shift assays are shown in Figure 2. The three loading control lanes contained the oligonucleotide added to varying amounts of purified NF-κB; band intensity in these lanes was related linearly to the amount of sample.

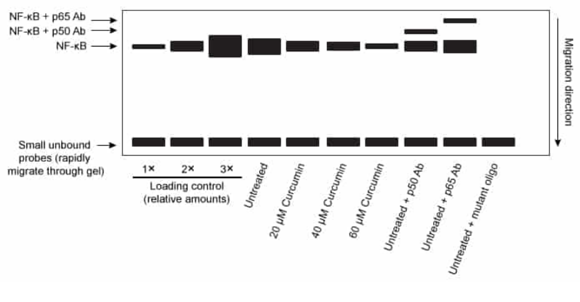


Figure 2 Electrophoretic mobility shift assay of NF-κB binding to DNA (Note: p50 Ab and p65 Ab refer to antibodies that bind the p50 and p65 subunits of NF-κB).

Table 1 quantifies the selected band intensities from several experiments.

Table 1 Quantification of Mean Band Intensities from Multiple Gels

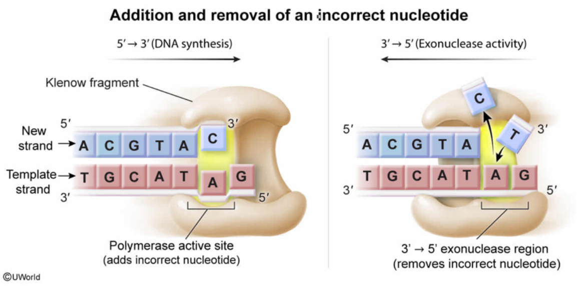
	1× Loading control	2× Loading control	3× Loading control	Untreated	20 μM curcumin	40 μM curcumin	60 μM curcumin
Relative band intensity	0.23	0.75	1.24	1	0.74	0.51	0.25

During nucleic acid synthesis, the Klenow fragment incorporates an incorrect nucleotide, which is subsequently removed. Compared with the action of the polymerase region of the Klenow fragment, its exonuclease region most likely acts:

- A. on the same nucleotide, but in the 5′ to 3′ direction. [14%]
- B. on the same nucleotide, but in the 3′ to 5′ direction. [63%]
- C. in the same direction, but on a nucleotide closer to the 5′ end of the strand. [12%]
- D. in the same direction, but on a nucleotide farther from the 5′ end of the strand. [10%]

Correct
63% answered correctly

Explanation:



DNA polymerases catalyze the accurate addition of nucleotides to growing strands of DNA in a 5′→3′ direction. To accomplish this, DNA polymerases possess both polymerase and exonuclease activities.

The passage describes the Klenow fragment of DNA polymerase I and asks where its exonuclease site acts relative to the site of action of the fragment's polymerase. The fact that each DNA strand is synthesized in a 5′→3′ direction means that the **polymerase** successively **adds** new **nucleotides to** the **3′ end** of the **new DNA strand**.

Exonucleases act at the ends of DNA strands, and a **3′→5′ exonuclease acts at** the **3′ end** (ie, the growing end) **of** a **new DNA strand**. In other words, a 3′→5′ exonuclease acts in the opposite direction of a polymerase to remove nucleotides that the polymerase recently added to the growing strand. Therefore, the **3′→5′ exonuclease acts** at the **same site as** the **polymerase but in the 3′→5′ direction**.

(Choice A) As stated in the passage, the Klenow fragment lacks the 5′→3′ exonuclease activity of the complete DNA polymerase from which it is isolated. Therefore, the Klenow fragment cannot act in the 5′→3′ direction (ie, removing nucleotides from the 5′ end).

(Choice C) Exonucleases act at the *ends* of DNA strands. Because DNA is synthesized in the 5′→3′ direction, acting on a nucleotide closer to the 5′ end of the strand than the last nucleotide added would mean acting on an interior nucleotide. Endonucleases, not exonucleases, perform this function.

(Choice D) Because the polymerase adds nucleotides in the 5′→3′ direction and the 3′→5′ exonuclease acts (by definition) at the 3′ end of the growing chain of nucleotides, no nucleotides are farther from the 5′ end of the strand than the one most recently added.

Educational objective:
DNA polymerases possess several different enzyme activities. A 3′→5′ exonuclease proofreads and helps make corrections at the 3′ end of a growing DNA strand.

Time spent:0 sec
Subject:
Biochemistry

QID:403428
System:
Carbohydrates, Nucleotides, and Lipids

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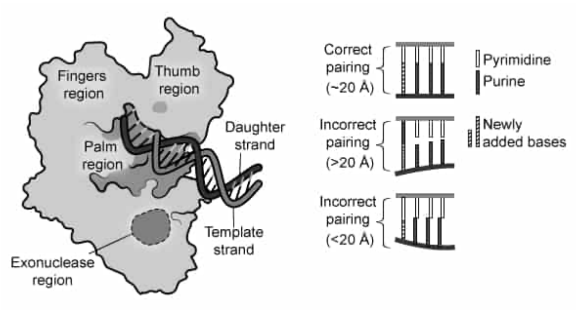

Figure 1 The Klenow fragment of DNA polymerase I (Note: The polymerase and exonuclease regions are separated by 35 angstroms.)

A team of researchers used the Klenow fragment to synthesize experimental oligonucleotides labelled with ^{32}P. These oligonucleotides contained either the binding sequence for a transcription factor called nuclear factor-kappa B (NF-κB) or a mutated sequence of the same length that lacks the binding site. The oligonucleotide probes were added to nuclei isolated from liver cells previously treated with the natural product curcumin or with a control solution.

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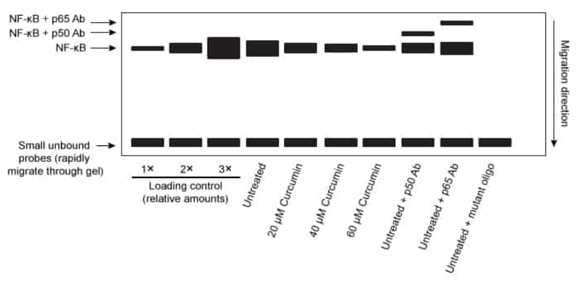

Figure 2 Electrophoretic mobility shift assay of NF-κB binding to DNA (Note: p50 Ab and p65 Ab refer to antibodies that bind the p50 and p65 subunits of NF-κB).

Table 1 quantifies the selected band intensities from several experiments.

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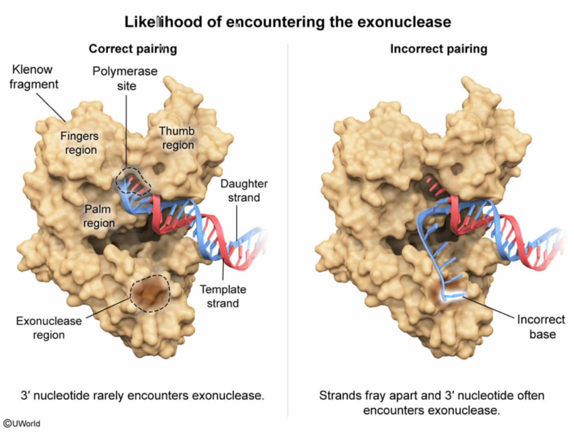
Relative band intensity	0.23	0.75	1.24	1	0.74	0.51	0.25

After removal of an incorrectly incorporated nucleotide by the Klenow fragment exonuclease, the correct nucleotide is incorporated. Compared to the incorrectly incorporated nucleotide, the chances of the correctly incorporated nucleotide encountering the exonuclease site of the Klenow fragment will most likely be:

- A. increased. [12%]
- B. decreased. [51%]
- C. approximately the same. [26%]
- D. approximately the same, but the encounter will occur more rapidly. [10%]

Correct
51% answered correctly

Explanation:



Among the several enzymes acting together at replication forks to synthesize a new DNA strand, DNA polymerase is the one that adds the new nucleotides to the 3′ ends of the growing DNA strands. The nearly identical size and shape of correct base pairs (ie, A with T, G with C) is optimal for interaction with the polymerization site, which may help prevent incorporation of incorrectly paired nucleotides into the growing DNA strand.

Although DNA polymerase is very accurate, it occasionally adds an incorrect nucleotide. When this occurs, DNA polymerase's 3′→5′ exonuclease typically removes the incorrect nucleotide before any more nucleotides are added by the polymerase.

The question asks about the relative chances of an incorrect nucleotide or its correct replacement encountering the exonuclease site of the Klenow fragment. As shown in Figure 1, **incorrect base pairing alters** the geometry of a new **DNA strand** and **directs** it on a new trajectory, often **toward the 3′→5′ exonuclease** site. **Replacement of** the incorrect nucleotide **with** the **correct one restores** the **normal geometry** of the strand, making it less likely to encounter the 3′→5′ exonuclease. Therefore, the **chances of** the **correctly incorporated nucleotide encountering** the **exonuclease** site will be **decreased** relative to those for the incorrectly incorporated nucleotide.

(Choice A) Because the geometry of an incorrectly paired base increases the likelihood of the 3′ nucleotide encountering the exonuclease site, correction of the incorrect pairing *decreases* the chance of the correctly paired 3′ nucleotide encountering the exonuclease site.

(Choices C and D) The chance of an incorrect 3′ nucleotide encountering the exonuclease site is greater than (*not* the same as) that of a correct one because of the change in the shape of the growing strand caused by incorrect base pairing.

Educational objective:
For a growing DNA strand to have the proper shape, it must have the correct base pairing. An incorrectly paired nucleotide alters the shape of a growing DNA strand in a way that increases the likelihood that the nucleotide will encounter the 3′→5′ exonuclease and be removed.

Time spent:0 sec	**QID**:403429
Subject: Biochemistry	**System:** Carbohydrates, Nucleotides, and Lipids

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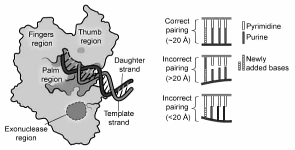

Figure 1 The Klenow fragment of DNA polymerase I (Note: The polymerase and exonuclease regions are separated by 35 angstroms.)

A team of researchers used the Klenow fragment to synthesize experimental oligonucleotides labelled with ^{32}P. These oligonucleotides contained either the binding sequence for a transcription factor called nuclear factor-kappa B (NF-κB) or a mutated sequence of the same length that lacks the binding site. The oligonucleotide probes were added to nuclei isolated from liver cells previously treated with the natural product curcumin or with a control solution.

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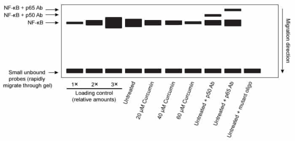

Figure 2 Electrophoretic mobility shift assay of NF-κB binding to DNA (Note: p50 Ab and p65 Ab refer to antibodies that bind the p50 and p65 subunits of NF-κB).

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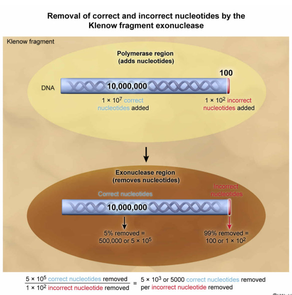
The Klenow fragment incorporates approximately 1×10^2 incorrect nucleotides for every 1×10^7 total nucleotides. If the exonuclease site removes 99% of the incorrect nucleotides while also removing 5% of the correctly incorporated nucleotides, which of the following statements describes nucleotide removal by the Klenow fragment?

- A. 1 correct nucleotide is removed for every 5000 incorrect nucleotides removed [31%]
- B. 100,000 correct nucleotides are removed for every 1 incorrect nucleotide removed [8%]
- C. 5000 incorrect nucleotides are removed for every 1 correct nucleotide removed [22%]
- ✔ D. 1 incorrect nucleotide is removed for every 5000 correct nucleotides removed [37%]

Correct
36% answered correctly

Explanation:



In addition to the polymerase activity for which DNA polymerases are named, they also possess the ability to remove incorrectly incorporated nucleotides via their exonuclease activity. The passage states that the Klenow fragment possesses exonuclease activity in the 3′→5′ direction.

The question asks for the statement that describes the removal of nucleotides by the Klenow fragment. Based on the information given, 1×10^2 (ie, 100) incorrect nucleotides are incorporated into a growing strand for every 10^7 (ie, 10 million) total nucleotides. Of the incorrect nucleotides, 99% (ie, 99) are subsequently removed.

Most of the nucleotides in the strand are correctly incorporated:

$$10^7 \text{ total bases } - 10^2 \text{ incorrect bases} = 10{,}000{,}000 \text{ total bases} - 100 \text{ incorrect bases} = 9{,}999{,}900 \text{ correct bases}$$

9,999,900 correct bases can be approximated as 10,000,000 (ie, 10^7) bases (in other words, the 100 incorrect bases are negligible for the calculation). The question states that 5% of **correctly incorporated** nucleotides are **removed** by the Klenow fragment exonuclease. The number of correct bases removed per 10^7 total bases can therefore be approximated as

$$5\% \times {\sim}10^7 \text{correct bases} \approx 500{,}000 \text{ correct bases removed}$$

The ratio of correct bases removed to incorrect bases removed is approximately

$$\frac{500{,}000 \text{ correct bases removed}}{99 \text{ incorrect bases removed}} \approx \frac{5000\cancel{00} \text{ correct bases removed}}{1\cancel{00} \text{ incorrect bases removed}} = \frac{5000 \text{ correct bases removed}}{1 \text{ incorrect bases removed}}$$

Consequently, approximately **1 incorrect nucleotide** is removed for every **5000 correct nucleotides** removed.

(Choices A and C) These are equivalent statements, both incorrectly asserting that the exonuclease removes far more total *incorrect* nucleotides than *correct* ones. Although the exonuclease removes a lower *percentage* of correct nucleotides than incorrect ones, the *total number* of correct nucleotides is so much larger than the total number of incorrect nucleotides that *more total* correct nucleotides are removed than incorrect ones.

(Choice B) For every 100,000 (ie, 10^5) correct nucleotides *added* by the Klenow fragment *polymerase*, approximately 1 incorrect nucleotide will be added (and subsequently removed). This is the relationship given in the question, not the relationship that was asked for.

Educational objective:
Incorrect nucleotides added to the 3′ end of a new DNA strand by DNA polymerase can be removed by the 3′→5′ exonuclease of DNA polymerase.

Time spent:0 sec
Subject:
Biochemistry

QID:403430
System:
Carbohydrates, Nucleotides, and Lipids

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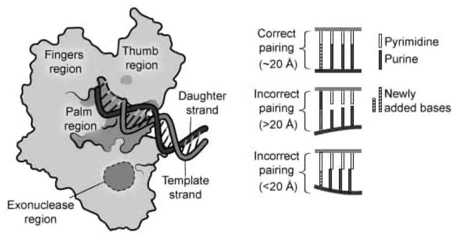

Figure 1 The Klenow fragment of DNA polymerase I (Note: The polymerase and exonuclease regions are separated by 35 angstroms.)

A team of researchers used the Klenow fragment to synthesize experimental oligonucleotides labelled with ^{32}P. These oligonucleotides contained either the binding sequence for a transcription factor called nuclear factor-kappa B (NF-κB) or a mutated sequence of the same length that lacks the binding site. The oligonucleotide probes were added to nuclei isolated from liver cells previously treated with the natural product curcumin or with a control solution.

After incubation, the samples underwent polyacrylamide gel electrophoresis. NF-κB proteins are dimers that may consist of various combinations of subunits, and antibodies that bind to specific NF-κB subunits were applied to some samples prior to electrophoresis. Typical band patterns and intensities from the electrophoretic mobility shift assays are shown in Figure 2. The three loading control lanes contained the oligonucleotide added to varying amounts of purified NF-κB; band intensity in these lanes was related linearly to the amount of sample.

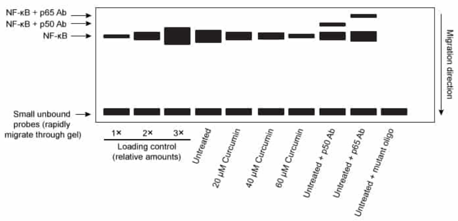

Figure 2 Electrophoretic mobility shift assay of NF-κB binding to DNA (Note: p50 Ab and p65 Ab refer to antibodies that bind the p50 and p65 subunits of NF-κB).

Table 1 quantifies the selected band intensities from several experiments.

Table 1 Quantification of Mean Band Intensities from Multiple Gels

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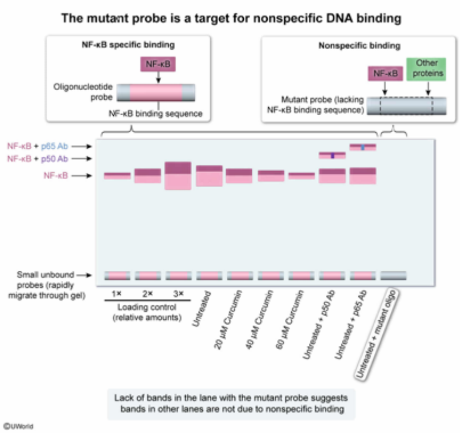
Relative band intensity	0.23	0.75	1.24	1	0.74	0.51	0.25

Given that the mutated oligonucleotide lacks the sequence needed to bind NF-κB, what was the most likely purpose of including the mutated oligonucleotide probe in the experiment? The mutated oligonucleotide:

- A. served as a positive control for measuring NF-κB activation. [25%]
- B. acts as a competitor with the canonical nucleotide for binding to NF-κB. [2%]
- ✔ C. served to confirm that DNA binding by NF-κB is specific to the proposed binding sequence. [69%]
- D. helped prevent excessive degradation of the canonical oligonucleotide by the exonucleases. [2%]

Correct
68% answered correctly

Explanation:



Transcription factors bind specific DNA sequences (or other DNA-bound proteins), and thereby regulate gene transcription. Transcription factor binding to DNA is sometimes assessed using short oligonucleotide probes that are in some way labeled (eg, radioactivity, fluorescence) to make them easily detectable.

In an electrophoretic mobility shift assay (EMSA), samples incubated with an oligonucleotide probe are separated on a native gel prior to subsequent detection. The migration of protein-DNA complexes is slowed relative to the migration of the smaller unbound probe.

The passage describes an experiment in which an EMSA was performed, and the question asks for the likely purpose of the mutated oligonucleotide probe lacking the consensus NF-κB binding site. The **mutated oligonucleotide probe** is a **target for nonspecific** NF-κB **binding** (ie, interactions between NF-κB and DNA of a sequence other than the binding sequence). Nonspecific binding should not occur if the radioactive probe is specific for NF-κB (ie, the mutated oligonucleotide probe serves as a negative control to verify that a result that should not occur, does not occur).

(Choice A) The researchers measured NF-κB activation by detecting the radioactivity from an oligonucleotide probe containing the NF-κB consensus binding site to which NF-κB has bound. Therefore, a positive control would be a sample known beforehand to produce a radioactive band corresponding to a complex of NF-κB bound to its consensus site, such as the bands in the loading control lanes. The mutant oligonucleotide cannot form such a complex.

(Choice B) Because it lacks the consensus sequence that NF-κB is known to bind, the mutated oligonucleotide is unlikely to be an effective competitor with the nucleotide that contains the consensus sequence.

(Choice D) Although the Klenow fragment possesses an exonuclease site, this site only removes nucleotides as they are being synthesized, not after replication is complete and the oligonucleotide is no longer associated with the polymerase.

Educational objective:
Mutations can influence interactions of other molecules with DNA. Negative and positive controls help verify that an experimental result is due to the independent and dependent variables under study rather than other factors. A negative control is an experimental condition in which no response to or signal from an experimental intervention is expected.

Time spent:0 sec
Subject:
Biochemistry

QID:403431
System:
Carbohydrates, Nucleotides, and Lipids

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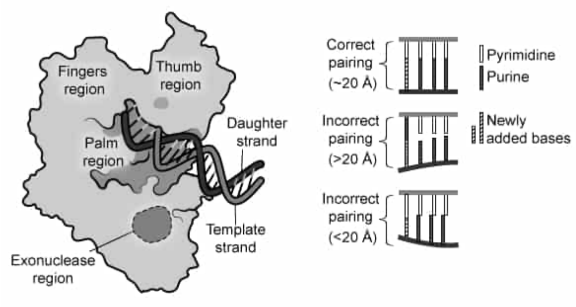

Figure 1 The Klenow fragment of DNA polymerase I (Note: The polymerase and exonuclease regions are separated by 35 angstroms.)

A team of researchers used the Klenow fragment to synthesize experimental oligonucleotides labelled with ^{32}P. These oligonucleotides contained either the binding sequence for a transcription factor called nuclear factor-kappa B (NF-κB) or a mutated sequence of the same length that lacks the binding site. The oligonucleotide probes were added to nuclei isolated from liver cells previously treated with the natural product curcumin or with a control solution.

After incubation, the samples underwent polyacrylamide gel electrophoresis. NF-κB proteins are dimers that may consist of various combinations of subunits, and antibodies that bind to specific NF-κB subunits were applied to some samples prior to electrophoresis. Typical band patterns and intensities from the electrophoretic mobility shift assays are shown in Figure 2. The three loading control lanes contained the oligonucleotide added to varying amounts of purified NF-κB; band intensity in these lanes was related linearly to the amount of sample.

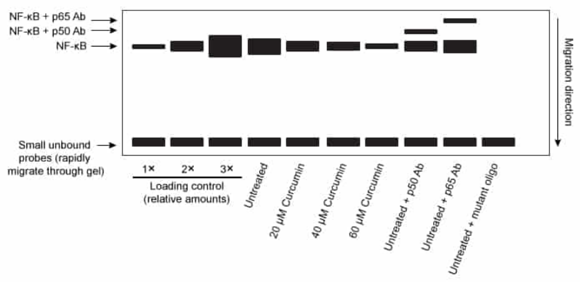

Figure 2 Electrophoretic mobility shift assay of NF-κB binding to DNA (Note: p50 Ab and p65 Ab refer to antibodies that bind the p50 and p65 subunits of NF-κB).

Table 1 quantifies the selected band intensities from several experiments.

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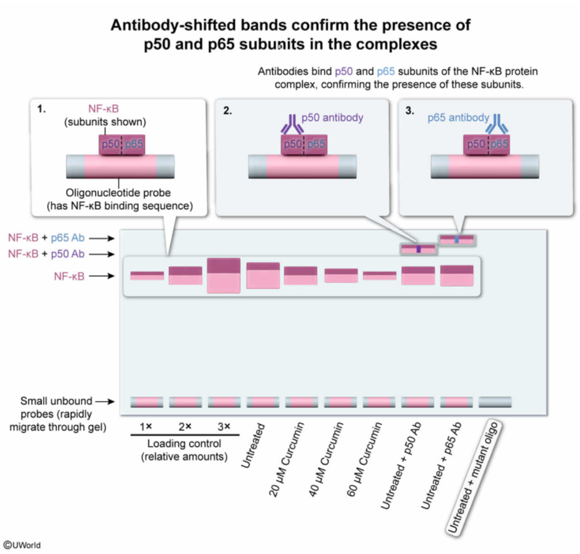
Why were the positions of the oligonucleotide complexes with NF-κB measured in both the presence and absence of antibodies to NF-κB subunits?

- A. To confirm that autoantibodies to the NF-κB subunits could be part of the complexes [9%]
- B. To determine whether the presence of additional proteins in the complexes would affect the band intensity [28%]
- C. To determine the effect of known molecular weight changes on complex position in the gel [9%]
- ✔ D. To confirm the presence or absence of specific subunits in the complexes [52%]

Correct
53% answered correctly

Explanation:



Transcription factors bind to gene regulatory sequences, thereby regulating transcription of the gene by RNA polymerases.

The passage describes an experiment in which NF-κB binding to radioactive oligonucleotide probes was assessed. The question asks why the positions of oligonucleotide complexes with NF-κB were measured in both the presence and absence of antibodies to NF-κB subunits. Antibodies are proteins that bind to portions of biological molecules known as epitopes.

Typically, antibody-epitope interactions are highly specific, meaning the antibody will not bind to other molecules. As such, each antibody added is expected to bind one and only one subunit of NF-κB. Antibody binding to NF-κB subunits alters the sizes of complexes containing those subunits. Therefore, **bands that shift in** the **presence of** an **antibody** (as shown in Figure 2) **confirm** the **presence of** the corresponding **subunit in** the **complex**.

(Choice A) Autoantibodies are antibodies produced in the body that recognize other molecules produced in the body. Transcription factor autoantibodies exist, but the antibodies in this experiment would detect NF-κB itself, not other antibodies bound to NF-κB.

(Choice B) The bands in the gel are detectable due to the radioactivity of the oligonucleotide probe in the complexes. Therefore, adding an additional protein (ie, the antibody) to the complex may change the *position* of complexes to which the antibody binds but not the inherent radioactive signal intensity from the labeled probe in the complex.

(Choice C) The binding of an antibody adds size to a complex. However, the purpose of the antibodies was to determine whether the relevant subunits were present, *as measured* by the occurrence of an antibody-induced change in mobility (not merely to quantify how mobility changed).

Educational objective:
Antibody binding shifts the migration of a molecule in a gel, thereby confirming the presence of the molecule targeted by that antibody.

Time spent:0 sec
Subject:
Biochemistry

QID:403432
System:
Carbohydrates, Nucleotides, and Lipids

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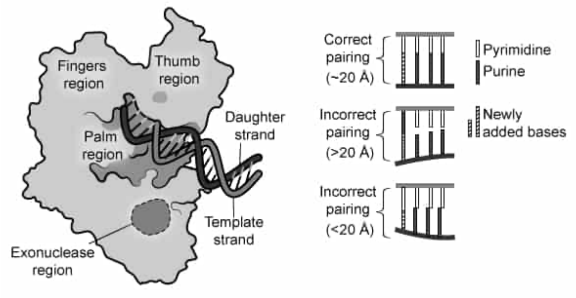

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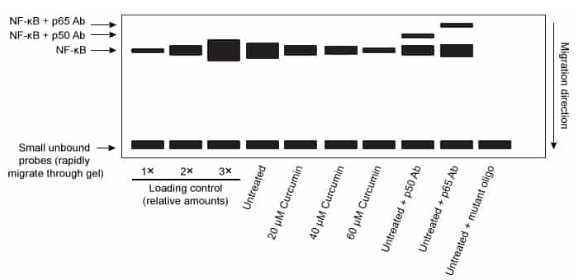

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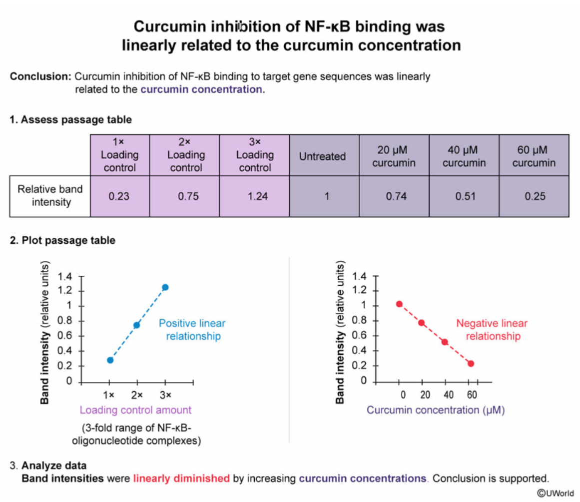
The researchers concluded that curcumin inhibition of NF-κB binding to target gene sequences was linearly related to the curcumin concentration. Do the data support the researchers' conclusion?

- A. No; the band intensities were related to the curcumin concentrations but not linearly. [9%]
- B. No; the bands corresponding to curcumin-treated samples were outside the linear range. [3%]
- ✔ C. Yes; the band intensities were linearly diminished by increasing curcumin concentrations. [83%]
- D. Yes; the band intensities were linearly diminished by decreasing curcumin concentrations. [3%]

Correct
83% answered correctly

Explanation:



Experiments often attempt to measure quantitative relationships between variables. However, depending on the methods used for quantitation, relationships between variables as well as between measured and actual values may not be straightforward. For example, actual values below a method's detection limit have the same measured value or description (eg, "0" or "undetectable"). In addition, the actual relationship between some variables may be nonlinear (eg, hyperbolic) over part or all the ranges studied in an experiment.

To better interpret the meaning of the values measured in experiments, scientists analyze standards for which actual values for a certain characteristic (eg, concentration, amount, radioactivity) are known. Such standards allow the relationship between characteristics to be determined. For example, a standard curve can establish the relationship between known concentrations of a molecule and the measured absorbance.

The passage presents results from an experiment in which NF-κB binding to DNA was studied, and the question asks if the data support the conclusion that NF-κB binding was linearly related to curcumin concentration. The **loading controls established** a **range** over which **band intensities** are **linearly related to NF-κB binding**; the other lanes show **band intensities within this range**. Table 1 shows that for every 20 μM of curcumin added, band intensity decreases by approximately 0.25 units. Therefore, the decrease in intensity is directly proportional **to increasing curcumin concentration** (ie, it is linear), so the **data support the** researchers' **conclusion**.

(Choice A) The band intensities *were* linearly related to curcumin concentration.

(Choice B) The mean band intensities for the curcumin-treated samples (as well as the mean value for the untreated sample) ranged from 0.25–1.00, which *is* within the range (0.23–1.24) over which band intensity varied linearly with NF-κB binding in the loading control lanes.

(Choice D) The band intensities were linearly diminished with increasing, *not* decreasing, curcumin concentrations.

Educational objective:
Standards can be used to establish the relationship between variables over a particular range, which can be used to better interpret measured values in experimental samples.

Time spent:0 sec
Subject:
Biochemistry

QID:403433
System:
Carbohydrates, Nucleotides, and Lipids